MAT 504 — Computer Modelling in Materials Science

Week 2 Hands-on

From Boltzmann Weights to Metropolis Sampling

Objective.
In this exercise, we will start from a simple potential-energy model, use it to define the Boltzmann probability of different configurations, generate representative configurations with Monte Carlo sampling, and finally estimate equilibrium properties from averages over the sampled configurations. We will implement that logic for a simple one-dimensional double-well potential. The system is deliberately simple so that we can focus on the sampling ideas.
By the end of the session, you should be able to:

- construct a canonical Boltzmann probability density from a potential-energy function;
- implement the essential steps of the Metropolis algorithm;
- explain accepted downhill and uphill moves;
- distinguish equilibration (burn-in) from production sampling;
- estimate simple equilibrium observables from sampled configurations;
- compare a sampled distribution with the target Boltzmann distribution;
- explain why a high acceptance rate does not necessarily imply efficient sampling.

Computational setup

We will use Python with

```
import numpy as np
import matplotlib.pyplot as plt
```

A starter Jupyter notebook is provided. Complete the indicated cells and answer the short questions in the notebook.

We use reduced units with

$$k_B = 1.$$

The temperature therefore appears directly as T in the Boltzmann factor.

Important interpretation. The variable x in this exercise is a one-dimensional *configuration coordinate*. The Monte Carlo step number is an algorithmic index, not physical time. This is an equilibrium-sampling exercise, not a dynamics simulation.

Part 1 — Potential energy and the target distribution

We will use the double-well potential

$$U(x) = \left(x^2 - 1\right)^2 .$$

It has two equivalent low-energy basins separated by an energy barrier.

For the canonical ensemble, the target equilibrium probability density is

$$p_T(x) = \frac{1}{Z} \exp\left[-\frac{U(x)}{T}\right],$$

where, because x is continuous,

$$Z = \int \exp\left[-\frac{U(x)}{T}\right] dx.$$

We will evaluate and normalize this density numerically on a grid.

Use

$$T = 0.5$$

and a configuration-space grid such as

```
x_grid = np.linspace(-2.5, 2.5, 2001)
```

Task 1.1 — Plot $U(x)$ and the Boltzmann density

Complete the supplied code to:

1. evaluate $U(x)$ on the grid;
2. calculate the unnormalized Boltzmann weights;
3. normalize them numerically to obtain $p_T(x)$;
4. plot $U(x)$ and $p_T(x)$.

Mark the two minima and the barrier near $x = 0$ on the energy plot.

Where is the equilibrium probability density largest, and why?
Why is the equilibrium distribution **not** concentrated only at the two exact minima of $U(x)$?

Part 2 — Build a Metropolis sampler

Starting from a current state x, propose a trial state

$$x' = x + \xi,$$

where

$$\xi \sim \text{Uniform}(-\delta, \delta).$$

This is a symmetric proposal distribution.

The energy change is

$$\Delta U = U(x') - U(x),$$

and the Metropolis acceptance probability is

$$p_{\text{acc}} = \min\left[1, \exp\left(-\frac{\Delta U}{T}\right)\right].$$

Use the following baseline parameters:

$$N_{\text{steps}} = 20000, \qquad T = 0.5, \qquad \delta = 1.0, \qquad x_0 = 2.5.$$

A fixed random seed is supplied in the notebook so that everyone can reproduce the same result during class.

Task 2.1 — Complete the Metropolis loop

Complete the missing lines in the supplied function. Your implementation should:

1. propose a trial state;
2. calculate ΔU;
3. accept all downhill moves;
4. accept uphill moves with the Metropolis probability;
5. retain the current state when a proposal is rejected.

After running the sampler, report the acceptance fraction.

What happens to the Markov chain when a proposed move is rejected?
Why is occasional acceptance of an uphill move essential for finite-temperature equilibrium sampling rather than energy minimization?

Part 3 — Equilibration and ensemble averages

The initial configuration $x_0 = 2.5$ is deliberately not representative of equilibrium. The early part of the Markov chain is therefore an equilibration or *burn-in* period.

Task 3.1 — Inspect the trajectory

Plot, as functions of Monte Carlo step:

- the sampled coordinate x_n;
- the corresponding energy $U(x_n)$.

Inspect at least the first 3000 steps. For this exercise, use

$$N_{\text{burn}} = 1000$$

and define all later samples as the production region.

Task 3.2 — Estimate two observables

From the production samples, estimate

$$\langle x^2 \rangle$$

and the right-basin probability

$$P(x > 0).$$

The second quantity can be estimated by the fraction of production samples for which $x > 0$.

Also plot the running estimate of $P(x > 0)$ over the trajectory.

Why should the initial burn-in samples normally be excluded from equilibrium averages?
For this symmetric potential, what value should $P(x > 0)$ approach with sufficiently good sampling? Does your result behave as expected?

Part 4 — Does Metropolis reproduce the Boltzmann distribution?

This is the central validation step.

Task 4.1 — Compare sampled and target distributions

Using only production samples:

1. plot a normalized histogram of the sampled x values;
2. overlay the target Boltzmann density $p_T(x)$ calculated in Part 1.

The histogram and the target density should use the same horizontal axis.

Does the sampled histogram reproduce the main features of the target Boltzmann distribution? If the match is not exact, give one statistical reason why perfect agreement should not be expected from a finite trajectory.

Part 5 — Acceptance is not the same as efficient sampling

We now compare two proposal sizes at the same temperature:

$$\delta_{\text{small}} = 0.05, \qquad \delta_{\text{balanced}} = 1.0.$$

Use the same number of Monte Carlo steps and the same starting configuration for both runs.

Task 5.1 — Compare exploration

For each proposal size:

1. calculate the acceptance fraction;
2. plot the production trajectory x_n;
3. note whether both wells are explored repeatedly.

You should observe that the very small proposal size gives a high acceptance rate but slow exploration.

Task 5.2 — Compare autocorrelation of an observable

The notebook provides a helper function for the autocorrelation of an observable. Here we use the observable

$$A(x) = x.$$

The autocorrelation $C_x(k)$ measures how strongly the value of x at one Monte Carlo step is related to its value k steps later.

Plot $C_x(k)$ for the two proposal sizes on the same axes.

Which proposal size gives more efficient sampling in this comparison?
Your answer must use all three ideas:

- acceptance rate;
- exploration of the two wells;
- decay of the autocorrelation function.

Why would choosing the run with the highest acceptance rate be misleading?

Final question — What did the algorithm accomplish?

Complete the conceptual chain

$$\boxed{U(x) \longrightarrow p_T(x) \longrightarrow \text{Metropolis chain} \longrightarrow \text{production samples} \longrightarrow \langle A \rangle}.$$

In 3–4 sentences, explain the role of each step in this sequence.
In particular, distinguish between:

- the potential-energy model;
- the target equilibrium distribution;
- the sampling algorithm;
- the observable average estimated from the samples.

Before you leave

Make sure your notebook contains:

1. plots of the double-well potential and the target Boltzmann probability density;
2. a working Metropolis sampler;
3. the acceptance fraction for the reference run;
4. trajectory and running-average plots used to examine equilibration and define the production region;
5. estimates of $\langle x^2 \rangle$ and $P(x > 0)$ from the production samples;
6. a histogram of the production samples overlaid with the target Boltzmann density;
7. a comparison of small and balanced proposal step sizes, including their acceptance fractions and exploration behavior;
8. a comparison of the autocorrelation of x for the two proposal step sizes;
9. your written answers to Questions 1–5 and the final question.

Keep this notebook. Homework 2 will build directly on the Metropolis sampler and diagnostics developed today, extending them to investigate temperature and proposal-size effects more systematically.